

swtpcsim — CP/68 on the SWTPC 6800

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The built-in `swtpc` machine is a Motorola 6800 with SWTBUG on its MP-S console and a DC-4 floppy controller (`dc4`). `cp68.toml` is that same machine with a **CP/68** boot disk mounted on drive 0, so it comes up at SWTBUG's `$` prompt ready to boot:

```
cd examples/swtpc/cp68
swtpcsim cp68.toml
$                               (SWTBUG prompt)
D                               (SWTBUG disk boot)
```

`D` is SWTBUG's disk-boot command: it loads CP/68's cold-loader from track 0 and jumps into it. CP/68 signs on and drops to its `.` prompt:

```
HEMENWAY ASSOCIATES CP/68-1.0
.
```

From there the usual CP/68 commands work.

The disk

`CP68.DSK` is TSC's **CP/68 1.0**, single-density: $35 \text{ tracks} \times 18 \text{ sectors} \times 128 \text{ bytes} = 80,640 \text{ bytes}$. Its track 0 is laid out `0,1,2,4..18` (sector ID 0 first, no sector 3) — unlike FLEX, whose every track starts at sector 1. The DC-4 board probes this geometry from the image `size` and stamps the track-0 sector-ID map on mount, so nothing in the `.toml` configures tracks or sides. `docs/sources.md` records the provenance and the exact SHA-256.

Read-only by default

Drive 0 is mounted `writprotect = true`, so looking around CP/68 can never dirty the image — a fresh clone stays a fresh clone, and there is no undo to need. To save files back, drop the `writprotect` line from the drive in `cp68.toml`, or copy the `.DSK` somewhere writable first.

Where the machine comes from

This file is the shipped `machines/swtpc.toml` with one disk mounted — see that file for why each board and address is what it is. For the sibling FLEX boot examples, see `examples/swtpc/flex/`.